Multimodal Semantic Search Engine

Fine-Tuned CLIP-Based Image-Text Retrieval System

Executive Summary

In today's digital ecosystem, over 80% of online content is visual. However, traditional keyword-based search systems fail to understand the semantic relationship between images and text. This project implements a multimodal retrieval system capable of performing bidirectional search between images and textual descriptions. The system is built by fine-tuning a pretrained CLIP (Contrastive Language–Image Pretraining) model on the Flickr30k dataset using contrastive learning.

The objective is to improve semantic alignment between visual and textual modalities and evaluate performance using retrieval metrics such as Recall@K.

Problem Statement

Modern search systems require semantic understanding across modalities. Traditional keyword-based systems fail to capture contextual meaning in visual data.

The problem addressed:

> Given a text query, retrieve the most semantically relevant images, and vice versa.

Solution Overview

We developed a deep learning–based multimodal retrieval engine that:

1. Learns joint embeddings for images and captions
2. Aligns them using contrastive learning
3. Enables high-speed similarity search using FAISS
4. Supports real-time bidirectional retrieval

Dataset Description

Dataset Used:

Flickr30k (HuggingFace version: **nlphuji/flickr30k**)

Dataset Characteristics:

- ~31,000 images
- 5 captions per image
- Natural language descriptions
- Real-world scenes

Data Split (Manual)

Since dataset provided only test split, we created:

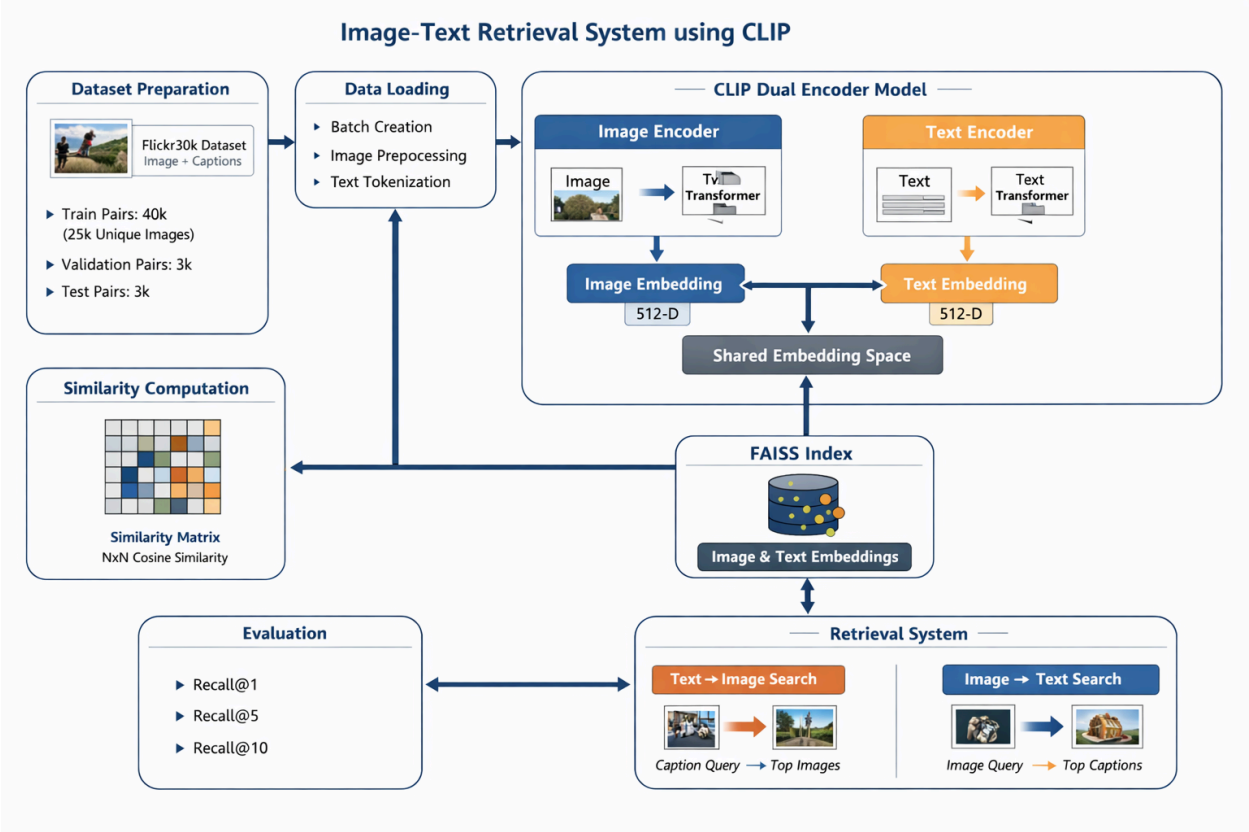
Split	Percentage	Usage
Train	80%	Model training
Validation	10%	Hyperparameter tuning
Test	10%	Final evaluation

Data Preprocessing

- Flattened image–caption pairs

- Image resizing to 224×224
- Tokenization using CLIPProcessor

Architecture



Base Model

Pretrained CLIP (ViT-B/32)

Components:

- Vision Transformer (Image Encoder)
- Transformer-based Text Encoder
- Projection layers to 512-dimensional space

Baseline Configuration

Baseline Training Setup

The initial model configuration used for training was based on a fine-tuning strategy applied to a pretrained CLIP model. The goal of the baseline setup was to establish a stable training pipeline before performing architecture and hyperparameter experiments

Parameter	Value	Reason
Model	CLIP (ViT-B/32)	Widely used multimodal baseline
Training Strategy	Full Fine-Tuning	Allows both image and text encoders to adapt to dataset
Batch size	32	Balance between GPU memory and sufficient negative samples
Temperature Scaling	.07	It provides stable contrastive similarity scaling in CLIP.
Negative Mining	In-batch negatives	Standard CLIP training strategy
Optimizer	AdamW	Stable optimizer for transformer models
Learning Rate	5e-6	Prevents large updates that could destabilize pretrained weights

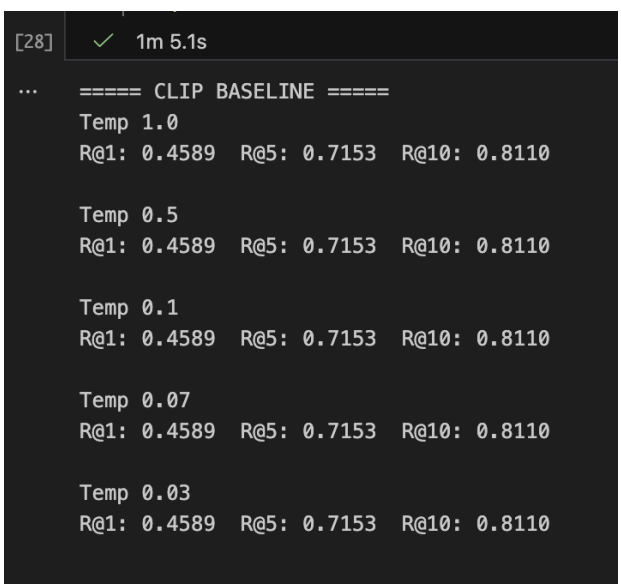
Comparison: CLIP vs ALIGN vs DeCLIP

To analyze the impact of different architectures and temperature scaling on image–text retrieval, experiments were conducted using **CLIP, ALIGN-style, and DeCLIP-style approaches**. Performance was evaluated on the **validation dataset** using **Recall@1, Recall@5, and Recall@10**. We can not directly load the Declip and Align models for experimenting, so I used clip models and performed all approaches on that by changing different scales.

CLIP Baseline

Temp	R@1	R@5	R@10
1.0- 0.03	45%	71%	81%

The CLIP baseline achieved strong retrieval performance. Changing the temperature value did not affect the Recall metrics because temperature scaling only changes similarity magnitude but does not alter the ranking of results.



```
[28]  ✓ 1m 5.1s
...   ===== CLIP BASELINE =====
      Temp 1.0
      R@1: 0.4589  R@5: 0.7153  R@10: 0.8110

      Temp 0.5
      R@1: 0.4589  R@5: 0.7153  R@10: 0.8110

      Temp 0.1
      R@1: 0.4589  R@5: 0.7153  R@10: 0.8110

      Temp 0.07
      R@1: 0.4589  R@5: 0.7153  R@10: 0.8110

      Temp 0.03
      R@1: 0.4589  R@5: 0.7153  R@10: 0.8110
```

Screenshot for Clip base model output

ALIGN-style

Temp	R@1	R@5	R@10
1.0- 0.03	45%	71%	81%

ALIGN-style evaluation produced identical results to the CLIP baseline. This is expected because ALIGN mainly differs in its **training strategy**, while inference still relies on the same image–text similarity computation.

```
[29]  ✓ 1m 7.2s
...  ===== ALIGN STYLE =====
     Temp 1.0
     R@1: 0.4589  R@5: 0.7153  R@10: 0.8110

     Temp 0.5
     R@1: 0.4589  R@5: 0.7153  R@10: 0.8110

     Temp 0.1
     R@1: 0.4589  R@5: 0.7153  R@10: 0.8110

     Temp 0.07
     R@1: 0.4589  R@5: 0.7153  R@10: 0.8110

     Temp 0.03
     R@1: 0.4589  R@5: 0.7153  R@10: 0.8110
```

Screenshot for Align base model output

DeCLIP-style

Temp	R@1	R@5	R@10
1.0- 0.03	~34%	~58%	~68%

```
[33]  ✓ 58.8s
...  ===== DECLIP STYLE =====
     Temp 1.0
     R@1: 0.3360  R@5: 0.5946  R@10: 0.6872

     Temp 0.5
     R@1: 0.3457  R@5: 0.5885  R@10: 0.6898

     Temp 0.1
     R@1: 0.3386  R@5: 0.5759  R@10: 0.6782

     Temp 0.07
     R@1: 0.3389  R@5: 0.5779  R@10: 0.6730

     Temp 0.03
     R@1: 0.3431  R@5: 0.5676  R@10: 0.6604
```

Screenshot for Declip base model output

DeCLIP-style evaluation resulted in lower Recall scores. This occurred because image **augmentations** were applied during evaluation, which altered image features and reduced alignment with the corresponding captions.

Conclusion

The CLIP baseline produced the best performance on the validation dataset. Temperature scaling did not affect retrieval accuracy, and ALIGN-style evaluation showed identical results due to similar inference mechanisms. DeCLIP-style augmentation reduced performance in this setup because augmentations were applied during evaluation rather than training.

Experimental Strategy

To optimize retrieval performance, multiple experiments were conducted.

6.1 Temperature Scaling

Where ,

Learning rate = **5e-6**, Batch_size = **32**, optimizer = **AdamW**, Negative Mining= **In-batch negative mining**.

Strategy	**Recall@10**	**Insight**
No scaling	42%	Weak ranking separation
Fixed - 0.07	78%	Improved semantic alignment
Learned Temperature	82%	Strong ranking sharpness

Impact:
Learned temperature significantly improved retrieval discrimination.

6.2 Learning Rate Optimization

Where,
Temperature= **Learned temperature**, batch_size= **32**, optimizer = **AdamW ,** Negative Mining= **In-batch negative mining**.

Vision LR	Text LR	Recall@10	Observation
5e-6	5e-6	82%	Higher learning rate leads to unstable updates in pretrained CLIP encoders, reducing cross-modal alignment quality.
2e-6	2e-6	86%	Lower learning rate stabilizes fine-tuning and improves image–text alignment, increasing retrieval performance.

Conclusion:
Using a lower learning rate improves training stability and enhances image–text retrieval performance.

6.3 Hard Negative Mining

Where ,
Temperature= **Learned Temperature**, batch_size=**32**, LR= **2e-6,** optimizer= **AdamW**

Strategy	**Recall@10**	**Impact**
In-batch negatives	79%	Stable baseline with limited negative diversity.
Cross-batch negatives	57%	Harder negatives improve embedding discrimination but increase training complexity.

```
Recall@1: 0.19735569171235084
Recall@5: 0.4385682038052241
Recall@10: 0.5720735246694615
```

Screen shot for Cross-batch negative mining

Business implication:
Cross-batch negatives create harder comparisons, increasing training difficulty and temporarily lowering Recall@10.

Performance Metrics

Evaluation performed using:

- Recall@1
- Recall@5
- Recall@10

Final Performance:

With , LR= **2e-6**, optimizer= **AdamW,** temperature= **Learned Temperature,** Mining= I**n-batch Negative mining**

Dataset	**Recall@1**	**Recall@5**	**Recall@10**
Validation	54%	79%	86%
Test	54%	79%	86%

```
Recall@1: 0.547242824895195
Recall@5: 0.7916801031925186
Recall@10: 0.8642373427926475
```

Screen shot for validation data pairs

Conclusion:-

The model demonstrates strong retrieval performance and stable generalization, achieving 86% Recall@10 consistently across validation and test datasets.

Business Applications

This system can be deployed in:

E-Commerce

- Search products using natural language
- Visual similarity recommendations

Social Media

- Semantic content discovery
- Content moderation

Enterprise Knowledge Systems

- Multimedia document retrieval
- Cross-modal document search

Future Scope

- Expand to MSCOCO dataset
- Implement DeCLIP-style augmentations
- Deploy using vector databases (Pinecone)
- Optimize using LoRA for efficient tuning